

Slope Interpretations in the Presence of Logs

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1 Log-Linear Model

1.1 Model Form

$$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 X$$

1.2 Derivation

Taking the exponential of both sides:

$$\hat{Y} = e^{\hat{\beta}_0 + \hat{\beta}_1 X} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 X}$$

Taking the derivative with respect to X :

$$\frac{d\hat{Y}}{dX} = e^{\hat{\beta}_0} \cdot \hat{\beta}_1 e^{\hat{\beta}_1 X} = \hat{\beta}_1 e^{\hat{\beta}_0 + \hat{\beta}_1 X} = \hat{\beta}_1 \hat{Y}$$

Rewriting as a relative change:

$$\frac{d\hat{Y}}{dX} = \hat{\beta}_1 \hat{Y} \quad \Rightarrow \quad \frac{d\hat{Y}/dX}{\hat{Y}} = \hat{\beta}_1$$

For small changes ΔX , the relative change in $\hat{Y}$ is approximately:

$$\frac{\Delta \hat{Y}}{\hat{Y}} \approx \hat{\beta}_1 \Delta X$$

Interpretation: When X increases by 1 unit, $\hat{Y}$ increases by approximately $\hat{\beta}_1 \times 100\%$.

2 Linear-Log Model

2.1 Model Form

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$$

2.2 Derivation

Taking the derivative with respect to X :

$$\frac{d\hat{Y}}{dX} = \hat{\beta}_1 \cdot \frac{1}{X} = \frac{\hat{\beta}_1}{X}$$

Rewriting in terms of relative change in X :

$$\frac{d\hat{Y}}{dX} = \frac{\hat{\beta}_1}{X} \quad \Rightarrow \quad d\hat{Y} = \hat{\beta}_1 \frac{dX}{X}$$

For small changes, $\frac{dX}{X} \approx \frac{\Delta X}{X}$ (the relative change in X):

$$\Delta \hat{Y} \approx \hat{\beta}_1 \cdot \frac{\Delta X}{X}$$

Interpretation: When X increases by 1% (i.e., $\frac{\Delta X}{X} = 0.01$), $\hat{Y}$ increases by approximately $0.01\hat{\beta}_1$ units.

3 Log-Log Model

3.1 Model Form

$$\log(\hat{Y}) = \hat{\beta}_0 + \hat{\beta}_1 \log(X)$$

3.2 Derivation

Taking the exponential of both sides:

$$\hat{Y} = e^{\hat{\beta}_0 + \hat{\beta}_1 \log(X)} = e^{\hat{\beta}_0} e^{\hat{\beta}_1 \log(X)} = e^{\hat{\beta}_0} X^{\hat{\beta}_1}$$

Taking the derivative with respect to X :

$$\frac{d\hat{Y}}{dX} = e^{\hat{\beta}_0} \cdot \hat{\beta}_1 X^{\hat{\beta}_1 - 1} = \hat{\beta}_1 \cdot \frac{e^{\hat{\beta}_0} X^{\hat{\beta}_1}}{X} = \hat{\beta}_1 \cdot \frac{\hat{Y}}{X}$$

Rewriting as a relative change:

$$\frac{d\hat{Y}}{dX} = \hat{\beta}_1 \frac{\hat{Y}}{X} \Rightarrow \frac{d\hat{Y}/dX}{\hat{Y}} = \frac{\hat{\beta}_1}{X}$$

Multiplying both sides by dX :

$$\frac{d\hat{Y}}{\hat{Y}} = \hat{\beta}_1 \frac{dX}{X}$$

For small changes:

$$\frac{\Delta \hat{Y}}{\hat{Y}} \approx \hat{\beta}_1 \frac{\Delta X}{X}$$

Interpretation: When X increases by 1% (i.e., $\frac{\Delta X}{X} = 0.01$), $\hat{Y}$ increases by approximately $\hat{\beta}_1 \times 1\% = \hat{\beta}_1\%$.

4 Summary

- **Log-linear:** $\frac{d\hat{Y}/dX}{\hat{Y}} = \hat{\beta}_1 \rightarrow 1$ unit increase in $X \rightarrow \hat{\beta}_1 \times 100\%$ increase in $\hat{Y}$
- **Linear-log:** $\frac{d\hat{Y}}{dX} = \frac{\hat{\beta}_1}{X} \rightarrow 1\%$ increase in $X \rightarrow 0.01\hat{\beta}_1$ unit increase in $\hat{Y}$
- **Log-log:** $\frac{d\hat{Y}/\hat{Y}}{dX/X} = \hat{\beta}_1 \rightarrow 1\%$ increase in $X \rightarrow \hat{\beta}_1\%$ increase in $\hat{Y}$ (elasticity)